



PLUMBING CD SUBMISSION 100%
08-15-2022

APPROVED BY	DATE	SPONSOR	DATE	MAINTENANCE ENG. OFF.	DATE	ENVIRONMENTAL MANAGEMENT	DATE	SAFETY	DATE	SHED - NIST FIRE MARSHAL/AHJ	DATE	FACILITIES ENGR. OFF.	DATE	VAY	AA	08/15/2022	Project Number	SH. NO. ____ OF ____	DWG. P001
																DATE	PROJECT / W.O. #		

PLUMBING GENERAL CONDITIONS

1. THE INTENT OF THESE DRAWINGS IS TO PROVIDE COMPLETE AND PROPERLY FUNCTIONING PLUMBING SYSTEMS. PROVIDE ALL LABOR AND MATERIAL NECESSARY TO ACHIEVE SUCH ENDS. CONTRACTOR IS OBLIGATED TO EXAMINE PLANS. ANY OBSERVED FAULTS OR AMBIGUITY IN THESE PLANS SHALL BE CALLED TO THE ATTENTION OF THE OWNER'S REPRESENTATIVE IMMEDIATELY. SO THAT THE MATTER MAY BE RESOLVED PRIOR TO SUBMISSION OF BIDS. BY SUBMISSION OF BID, THE CONTRACTOR SHALL ACKNOWLEDGE ACCEPTANCE OF THESE PLANS AS AN ADEQUATE DEFINITION OF THE SCOPE OF WORK AND EXTRA COST CLAIMS BASED ON INADEQUACY OF PLANS WILL NOT BE CONSIDERED.
2. ALL WORK ON THIS PROJECT SHALL BE INSTALLED IN ACCORDANCE WITH APPLICABLE CODES AND REGULATIONS.
3. THE PLUMBING CONTRACTOR SHALL COORDINATE THE INSTALLATION OF PLUMBING WORK WITH OTHER TRADES. MINOR DEVIATIONS FROM THE PLANS MAY BE MADE TO AVOID MINOR CONFLICTS. WHEN MAJOR CONFLICTS ARE APPARENT, THE ENGINEER SHALL BE ADVISED IMMEDIATELY, AND AFFECTED WORK SHALL NOT BE INSTALLED UNTIL THE CONFLICT HAS BEEN RESOLVED.
4. CONTRACTOR SHALL OBTAIN AND PAY FOR PERMITS AND ARRANGE FOR INSPECTIONS BY LOCAL AUTHORITIES HAVING JURISDICTION.
5. PROVIDE OPENINGS IN BUILDING CONSTRUCTION FOR PASSAGE OF PIPING. DO NOT PENETRATE STRUCTURAL MEMBERS WITHOUT PRIOR APPROVAL OF ARCHITECT AND STRUCTURAL ENGINEER. VERIFY SLAB PENETRATION LOCATIONS PRIOR TO PROCEEDING WITH WORK IN ORDER TO LOCATE OBSTRUCTIONS EMBEDDED IN SLAB.
6. ALL NECESSARY ALLOWANCES AND PROVISIONS SHALL BE MADE BY THIS CONTRACTOR FOR BEAMS, COLUMNS OR OTHER OBSTRUCTIONS OF THE BUILDING OR THE WORK OF OTHER CONTRACTORS, WHETHER OR NOT SAME IS INDICATED. WHERE NECESSARY TO AVOID OBSTRUCTIONS THE DUCTS SHALL BE TRANSFORMED, DIVIDED, OFFSET, RAISED OR LOWERED WITH THE REQUIRED FREE AREA BEING MAINTAINED WHILE MAINTAINING DESIGNED CEILING HEIGHTS.
7. ALL ROTATING PLUMBING EQUIPMENT SHALL BE MOUNTED ON VIBRATION ISOLATION IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
8. INSTALL ALL EQUIPMENT IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.
9. CONTROLS AND ACCESSORIES FOR EQUIPMENT SHALL BE PROVIDED BY EQUIPMENT MANUFACTURER.
10. PLUMBING CONTRACTOR SHALL THOROUGHLY CLEAN HIS WORK AREA DAILY AND SHALL ALSO REMOVE ALL TRASH AFTER WORK COMPLETION.
11. THE WORK DETAILED ON THESE PLANS IS BASED ON PREVIOUSLY PREPARED BASE BUILDING DRAWINGS. IF EXISTING FIELD CONDITIONS ARE DIFFERENT THAN SHOWN, THE CONTRACTOR SHALL ADVISE THE ENGINEER OF DISCREPANCIES WHICH WILL AFFECT THE PROPOSED WORK PRIOR TO BEGINNING WORK.
12. SYMBOLS SHOWN ON SCHEDULES DEFINE TYPE OF EQUIPMENT ONLY. CONTRACTOR IS RESPONSIBLE FOR RESEARCHING DRAWINGS FOR EXACT QUANTITIES REQUIRED OF EACH TYPE.
13. CONTRACTOR SHALL VERIFY THAT ALL EQUIPMENT, PIPING, ETC., SHALL BE FREE FROM INTERFERENCE WITH OTHER DISCIPLINES. WHERE CONFLICTS OCCUR, CONTRACTOR SHALL IMMEDIATELY CONTACT THE CONSTRUCTION MANAGER. WHERE THE WORK OF VARIOUS TRADES WILL BE INSTALLED IN CLOSE PROXIMITY TO ONE ANOTHER, OR WHERE THERE IS EVIDENCE THAT THE WORK OF ONE TRADE WILL INTERFERE WITH WORK OF OTHER TRADES, THE CONTRACTOR SHALL ASSIST IN WORKING OUT SPACE CONDITIONS TO MAKE A SATISFACTORY ADJUSTMENT. IF THE CONTRACTOR ALLOWS ONE TRADE TO INSTALL HIS WORK BEFORE COORDINATING WITH WORK OF OTHER TRADES, THE CONTRACTOR SHALL MAKE NECESSARY CHANGES TO CORRECT THE CONDITION WITHOUT EXTRA CHARGE.
14. THE CONTRACTOR SHALL LOCATE ALL EQUIPMENT WHICH MUST BE SERVICED, OPERATED, MAINTAINED IN FULLY ACCESSIBLE POSITION. EQUIPMENT SHALL INCLUDE, BUT NOT BE LIMITED TO, VALVES, TRAPS, CLEANOUTS, MOTORS, CONTROLLERS, DRAIN POINTS, ETC. IF REQUIRED FOR ACCESSIBILITY, FURNISH ACCESS DOORS FOR THIS PURPOSE. MINOR DEVIATIONS FROM DRAWINGS MAY BE MADE. ALLOW FOR BETTER ACCESSIBILITY, AND ANY CHANGE FOR THAT PURPOSE SHALL BE APPROVED.
15. ALL EQUIPMENT WHICH REQUIRES INTERFACE WITH THE ELECTRICAL TRADE, SUCH AS CONTROLLING DEVICES, DISCONNECT SWITCHES, MOTORS, ETC., SHALL BE FULLY COORDINATED BETWEEN ALL TRADES.
16. ALL WORK TO BE GUARANTEED FOR ONE YEAR AGAINST LABOR AND MATERIALS. ANY DEFECTIVE MATERIALS OR WORKMANSHIP, AS WELL AS DAMAGE TO THE WORK OF ALL TRADES RESULTING FROM SAME, SHALL BE REPLACE OR REPAIRED AS DIRECTED FOR THE DURATION OF THE GUARANTEE PERIOD. TIME FOR THIS GUARANTEE SHALL BEGIN FROM THE DATE OF ACCEPTANCE OF THE COMPLETE WORK BY THE OWNER OR HIS APPOINTED REPRESENTATIVE. NOTE: THESE GUARANTEES SHALL BE SUBMITTED TO THE CO FOR RECORD PURPOSES.
17. AS USED IN THE DRAWINGS FOR PLUMBING WORK, CERTAIN NON-TECHNICAL WORDS SHALL BE UNDERSTOOD TO HAVE SPECIFIC MEANINGS AS FOLLOWS REGARDLESS OF INDICATIONS TO THE CONTRARY IN THE GENERAL CONDITION OR OTHER DOCUMENTS GOVERNING THE MECHANICAL WORK.
 - A. "FURNISH" - PURCHASE AND DELIVER TO THE PROJECT SITE COMPLETE WITH EVERY NECESSARY APPURTENANCE AND SUPPORT. ALL AS PART OF THE MECHANICAL WORK. PURCHASING SHALL INCLUDE PAYMENT OF ALL SALES TAXES AND OTHER SURCHARGES AS MAY BE REQUIRED TO ASSURE THAT PURCHASED ITEMS ARE FREE OF ALL LIENS, CLAIMS OR ENCUMBRANCES.
 - B. "INSTALL" - UNLOAD AT THE DELIVERY POINT AT THE SITE AND PERFORM EVERY OPERATION NECESSARY TO ESTABLISH SECURE MOUNTING AND CORRECT OPERATION AT THE PROPER LOCATION IN THE PROJECT, ALL AS PART OF THE MECHANICAL WORK.
 - C. "PROVIDE" - "FURNISH" AND "INSTALL".
 - D. "NEW" - MANUFACTURED WITHIN THE PAST TWO YEARS AND NEVER BEFORE USED.

PLUMBING GENERAL NOTES	
1.	BEFORE COMMENCEMENT OF WORK, CONTRACTOR SHALL VERIFY EXACT LOCATIONS, ELEVATIONS AND CHARACTERISTICS OF UTILITIES AND PIPING AND SHALL IMMEDIATELY NOTIFY THE ARCHITECT/ENGINEER OF ANY DISCREPANCIES.
2.	THE ELECTRICAL CONTRACTOR IS REQUIRED TO PROVIDE POWER FOR ALL PLUMBING EQUIPMENT, DURING BID PERIOD AND AGAIN PRIOR TO EQUIPMENT ORDERING. THE GENERAL CONTRACTOR AND THE PLUMBING CONTRACTOR SHALL COORDINATE WITH THE ELECTRICAL CONTRACTOR TO VERIFY THAT ALL EQUIPMENT ARE POWERED, AND PROPER VOLTAGE/WIRING/CAPACITY IS PROVIDED. IN THE EVENT OF REQUIREMENTS AMBIGUITY OR CONFLICT BETWEEN TRADES, THE CONTRACTOR IS REQUIRED TO GENERATE AN RFI ADDRESSING THE ISSUE AND REQUESTING CLARIFICATION. FAILURE TO DO SO WILL NEGATE HIS RIGHT FOR A CHANGE ORDER OR REQUEST FOR ADDITIONAL FUNDS DURING CONSTRUCTION RELATING TO THIS ISSUE.
3.	DRAWINGS ARE IN PART DIAGRAMMATIC, INTENDED TO CONVEY SCOPE OF WORK, AND TO INDICATE THE GENERAL LOCATIONS OF EQUIPMENT, PIPING, ETC. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND LAYOUT HIS WORK ACCORDING TO THE FOLLOWING GUIDELINES:
A.	CONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING THE EXACT LOCATIONS FOR EQUIPMENT AND ROUGH INS AND THE EXACT ROUTING OF PIPING PRIOR TO CONSTRUCTION AS TO BEST FIT THE LAYOUT OF THE WORK. COORDINATE FINAL LAYOUT WITH ALL TRADES.
B.	WHERE OFFSETS IN PIPING ARE REQUIRED TO COORDINATE WORK OF OTHER TRADES, STRUCTURE, PIPING, CONDUITS, DUCTWORK, ETC. OR TO MAINTAIN CEILING HEIGHTS, THEY SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
C.	UNLESS OTHERWISE NOTED, ALL PIPING TO BE ROUTED CONCEALED IN WALLS, CHASES OR ABOVE CEILING. WATER PIPING SHALL NOT BE ROUTED IN EXTERIOR WALLS. ROUTE ALL PIPING AS HIGH AS POSSIBLE AND ALONG WALLS TO MAXIMIZE SPACE AVAILABLE FOR OTHER TRADES.
D.	COORDINATE ROUTING OF PIPING TO MAINTAIN ACCESS TO FILTERS, MOTORS, ELECTRICAL EQUIPMENT, AND CONTROLS. IN NO CASE SHALL PIPING PASS DIRECTLY OVER ELECTRICAL PANELS OR DISCONNECTS OR RESTRICT ACCESS TO ANY ELECTRICAL EQUIPMENT INCLUDING JUNCTION BOXES.
E.	ALL EXISTING PIPING SHOWN IS INTENDED TO INDICATE APPROXIMATE SIZE, NUMBER AND LOCATION OF PIPING BRANCHES FOR BIDDING PURPOSES ONLY. CONTRACTOR TO VERIFY EXACT SIZE AND CONFIGURATION PRIOR TO CONSTRUCTION.
F.	VERIFY EXACT SIZE AND LOCATION OF ALL EXISTING SYSTEMS REQUIRING CONNECTION TO NEW PIPING PRIOR TO COMMENCING WORK.
4.	UNLESS OTHERWISE NOTED, SANITARY WASTE PIPING SHOWN IS BELOW FLOOR AND ALL OTHER PIPING IS OVERHEAD, ABOVE CEILING, DOMESTIC HOT, COLD AND RE-CIRC. WATER PIPING SHALL BE INSTALLED ON WARM SIDE OF INSULATION.
5.	ALL PLUMBING WORK SHALL BE DONE IN ACCORDANCE WITH 2018 INTERNATIONAL PLUMBING CODE (IPC) & ALL LOCAL AMENDMENTS, CODES AND REGULATIONS.
6.	DURING THE PROGRESS OF THE WORK, MAINTAIN AN ACCURATE RECORD OF ALL CHANGES MADE IN THE PLUMBING SYSTEMS. THE RECORD DRAWING SHALL SHOW CHANGES IN MANUFACTURER (WITH NUMBERS AND NAMES), MATERIALS, SIZES, LOCATING AND HOOK-UP POINTS. AS-BUILTS SHALL BE GIVEN TO [OWNER] AT COMPLETION OF JOB.
7.	UPON COMPLETION OF JOB, THIS CONTRACTOR SHALL INSPECT ALL EXPOSED PORTIONS OF THE PLUMBING INSTALLATION AND COMPLETELY REMOVE ALL EXPOSED LABELS, SOIL, MARKING AND FOREIGN MATERIAL EXCEPT PRODUCT LABELS AND THOSE REQUIRED BY LAW TO REMAIN.
8.	ALL INSTALLATION OF ACCESSIBLE PLUMBING FIXTURES SHALL BE PERFORMED IN ACCORDANCE WITH THE REQUIREMENTS OF THE STATE ACCESSIBILITY STANDARDS. REFER TO ARCHITECTURAL PLANS FOR SPECIFIC INSTALLATION REQUIREMENTS. LOCATE WATER CLOSET FLUSH VALVE LEVER BETWEEN TOILET AND LAVATORY. COORDINATE LOCATION OF FLUSH VALVE LEVER HANDLE WITH GRAB BAR. COORDINATE WITH ARCHITECTURAL DRAWINGS.
9.	ALL VENT AND FLUE OUTLETS SHALL BE 10'-0" MINIMUM HORIZONTALLY AND 3'-0" MINIMUM VERTICALLY FROM ANY FRESH AIR INTAKES.
10.	PROVIDE WATER HAMMER ARRESTORS AT ALL PLUMBING FIXTURES OR BATTERY OF FIXTURES WITH QUICK-CLOSING VALVES. WATER HAMMER ARRESTORS SHALL BE SIZED & INSTALLED PER STANDARDS. AIR CHAMBERS SHALL NOT BE CONSIDERED AN EQUAL TO WATER HAMMER ARRESTORS AND SHALL NOT BE INSTALLED.
11.	PROVIDE BALANCING VALVES WITH THREADED CONNECTIONS IN HOT WATER RETURN PIPING, AND BALANCE SYSTEM FOR PROPER OPERATION.
12.	COORDINATE WITH ARCHITECTURAL DRAWINGS BEFORE ROUGHING IN PLUMBING.
13.	ALL OPENINGS IN CEILINGS AND PLENUM WALLS FOR PLUMBING SHALL BE SEALED AIR TIGHT AND PROTECTED WITH FIRESTOP.
14.	SEE SITE PLAN FOR EXTENT OF ALL PIPING LEAVING AND ENTERING BUILDING.
15.	ALL HOSE BIBS SHALL BE MOUNTED 18" ABOVE FINISHED FLOOR, UNLESS OTHERWISE SPECIFIED.
16.	ALL WALL HYDRANTS SHALL BE MOUNTED 24" ABOVE FINISHED GRADE UNLESS OTHERWISE SPECIFIED.
17.	MAKE PROPER HW, CW, RE-CIRC., WASTE, AND VENT CONNECTIONS TO ALL FIXTURES AND EQUIPMENT EVEN THOUGH ALL BRANCH MAIN, ELBOWS AND CONNECTIONS ARE NOT SHOWN.
18.	CLEANOUTS SHALL BE PROVIDED NEAR BASE OF EACH VERTICAL WASTE OR SOLID STACK. PROVIDE 18" MINIMUM CLEARANCE FOR ACCESS.
19.	PROVIDE VALVES AND UNIONS WHEN CONNECTING PIPING TO EQUIPMENT. LOCATE VALVES FOR EASY ACCESS AND SERVICING OF EQUIPMENT. PROVIDE ACCESS PANELS TO ALL SHUT-OFF VALVES.
20.	PRESSURE TEST ALL NEW WATER PIPES, MINIMUM OF 4 HOURS BEFORE APPLYING PIPE INSULATION. ALL TESTS SHALL BE IN COMPLIANCE WITH REQUIREMENTS OF THE LOCAL WATER AUTHORITY.
21.	IN THE EVENT OF CONFLICT BETWEEN VARIOUS SECTIONS OF CONSTRUCTION DOCUMENTS, THE MOST STRINGENT OF THE OPTIONS SHALL APPLY FOR BIDDING PURPOSE. AFTER AWARD OF CONTRACT BUT BEFORE STARTING DESIGN/CONSTRUCTION, CONTRACTOR SHALL REQUEST CLARIFICATIONS WITH BID ON ANY CONFLICTING INFORMATION FROM THE OWNER.
22.	PROVIDE DIELECTRIC FITTINGS OTHER THAN DIELECTRIC UNIONS WHEN CONNECTING DISSIMILAR MATERIAL PIPING. DIELECTRIC UNIONS NOT PERMITTED.
23.	ALL CONCRETE FLOORS, SLABS, BRICK AND BLOCK WALL PENETRATIONS SHALL BE SLEEVED AND FIRE STOPPED.
24.	PROVIDE BALL VALVE FOR ALL CW AND HW BRANCH TAKE-OFFS.
25.	ALL EXTERIOR ROOF PENETRATIONS SHALL BE SEALED WEATHERTIGHT.
26.	SLOPES AND INVERT ELEVATIONS OF ALL INTERIOR PIPING SHALL BE ESTABLISHED BEFORE ANY PIPING IS INSTALLED IN ORDER THAT PROPER SLOPES WILL BE MAINTAINED. ALL PIPING SHALL BE LOCATED AND DETERMINED WHERE TO BE RUN TO AVOID CONFLICT WITH OTHER TRADES.
27.	ALL PIPING SHALL BE PROVIDED WITH SECTIONAL, SHUT-OFF VALVES AND ACCESS PANELS FOR ALL HIDDEN VALVES. PROVIDE ISOLATION VALVES FOR ALL PLUMBING FIXTURES.
28.	ALL WORK SHALL BE GUARANTEED AGAINST DEFECTS, LEAKS, LACK OF PROPER SYSTEM PERFORMANCE OR NON-OPERATION FOR A PERIOD OF ONE YEAR AFTER DATED OF FINAL ACCEPTANCE OF WORK BY OWNER.
29.	ALL WORK SHALL BE COORDINATED WITH ALL TRADES PRIOR TO INSTALLATION.
30.	ALL EXPOSED PIPING TO FIXTURES SHALL BE HEAVILY CHROME-PLATED.
31.	MAKE FINAL CONNECTIONS AFTER FIXTURES AND EQUIPMENT ARE SET IN PLACE. ALL REQUIRED TRIM, FAUCETS AND MINOR ITEMS NECESSARY FOR A COMPLETE FUNCTIONING INSTALLATION, SUCH AS STRAINERS, TRAPS, TAILPIECES, ESCUTCHEONS, ETC., SHALL BE FURNISHED AND INSTALLED BY THE CONTRACTOR IN ACCORDANCE WITH EQUIPMENT REQUIREMENTS. ALL FAUCETS TO BE HEAVILY CHROME-PLATED.
32.	SLEEVES THROUGH CMU BLOCKS WALLS AND/OR MASONRY FLOORS SHALL BE SCHEDULE 40 BLACK STEEL PIPE. SLEEVES THRU FLOORS SHALL STAND PROUD OF FLOOR SLAB BY 1/2". SLEEVES THROUGH SHEET ROCK WALL SHALL BE 1" GAUGE GALVANIZED SHEET METAL. THE SPACE BETWEEN THE PIPE AND SLEEVE SHALL BE PACKED WITH "3M" EXPANDABLE FOAM FIRE-STOP MATERIAL, TO ASSURE A RATING EQUAL TO THE PENETRATED WALL OR FLOOR.
33.	HANGERS SHALL BE OF SIMILAR MATERIAL AS PIPING TO AVOID GALVANIC ACTION IN ACCORDANCE WITH MSS-SP-69, AND LOCATED AT INTERVALS NOT TO EXCEED 6'-0" O.C. WHERE "HUBLESS" PIPE IS BEING USED. PROVIDE HANGERS ON EITHER SIDE OF THE JOINTS AS RECOMMENDED BY CSPI BULLETIN 310. FOR INSULATED PIPING, INSULATION SHALL EXTEND CONTINUOUSLY THRU HANGER WITH OUT INTERRUPTION. PROVIDE SADDLE TO PROTECT INSULATION FROM COMPRESSION.
34.	SOIL, WASTE AND VENT PIPING SHALL BE INSTALLED IN ACCORDANCE WITH CODE. HORIZONTAL DRAINAGE PIPING SHALL SLOPE PER 2018 IPC CODE, UNLESS OTHERWISE INDICATED. DOMESTIC WATER SHALL BE SLOPED TO DRAIN POINTS.
35.	SUBMIT OPERATING AND MAINTENANCE MANUALS AND WARRANTY CERTIFICATES TO OWNER.
36.	THE CONTRACTOR SHALL NOT CORE DRILL CONCRETE SLABS OR PRE-CAST CONCRETE PANELS OR DECK WITHOUT THE KNOWLEDGE AND WRITTEN CONSENT OF THE STRUCTURAL ENGINEER AND THE BUILDING OWNER AND SHALL X-RAY PRIOR TO ANY CORE DRILLING. X-RAYING AND CORE DRILLING SLABS SHALL BE SCHEDULED WITH THE BUILDING OWNER, INCLUDE COSTS OF X-RAYING IN BID.
37.	FINAL ACCEPTANCE OF PLUMBING SYSTEMS BY THE USER/OWNER, SHALL BE CONTINGENT TO COMPLIANCE WITH ALL COMMERCIAL REQUIREMENTS AND TESTS PRESCRIBED IN THE PROJECT DESIGN DOCUMENTS/SPECIFICATIONS AND APPROVAL BY THE C&A AGENT OF THE PROJECT.
38.	UPON COMPLETION OF THE PROJECT, ALL DOMESTIC WATER PIPING SHALL BE SANITIZED IN ACCORDANCE WITH THE REQUIREMENTS OF THE AUTHORITY HAVING JURISDICTION.
39.	WHEN NATURAL GAS IS AVAILABLE, USE NATURAL GAS RANGES/OVENS & FURNACES IN SINGLE FAMILY & MULTI-FAMILY BUILDINGS AS AN OPTION, IN-LIEU OF ELECTRIC.

ABBREVIATIONS		
ABV	ABOVE	
ABV	AREA DRAIN	
AFF	ABOVE FINISH FLOOR	
AFG	ABOVE FINISH GRADE	
AP	ACCESS PANEL	
APPROX	APPROXIMATE	
AV	ACID VENT	
AW	ACID WASTE	
BAS	BUILDING AUTOMATION SYSTEM	
BFF	BELOW FINISH FLOOR	
BFP	BACKFLOW PREVENTER	
BLDG	BUILDING	
BT	BATTITUB	
CA	COMPRESSED AIR	
CD	CONDENSATE	
CLG	CEILING	
CO	COMPANY, CLEANOUT	
CW	COLD WATER	
DFU	DRAINAGE FIXTURE UNITS	
DIA	DIAMETER	
DN	DOWN	
DS	DOWNSPOUT	
DWG	DRAWING	
CW	COLD WATER	
DISP	DISPOSER	
DWV	DRAINAGE WASTE & VENT	
EA	EACH	
ECO	EXTERIOR CLEANOUT	
ETR	EXISTING TO REMAIN	
EW	EMERGENCY EYEWASH	
ES	EMERGENCY SHOWER	
EWC	ELECTRIC WATER COOLER	
EWH	ELECTRIC WATER HEATER	
EK	EXISTING	
FD	FLOOR DRAIN	
FFE	FINISHED FLOOR ELEVATION	
FIXT	FIXTURE	
FLR	FLOOR	
FM	FORCED MAIN	
FS	FLOOR SINK	
G	NATURAL GAS	
GCO	GRADE CLEANOUT	
GD	GROUNDWATER DRAIN TILE	
GPF	GALLONS PER FLUSH	
GPM	GALLONS PER MINUTE	
GWH	GAS WATER HEATER	
HB	HOSE BIBB	
HW	HOT WATER	
HWR	HOT WATER RECIRCULATING	
I.E.	INVERT ELEVATION	
INDRECT	INDIRECT WASTE	
IPC	INTERNATIONAL PLUMBING CODE	
LAV	LAVATORY	
MAU	MAKE-UP AIR UNIT	
MAX	MAXIMUM	
MB	MOP BASIN	
MBH	BRITISH THERMAL UNITS (x1000)	
MIN	MINIMUM	
MISC	MISCELLANEOUS	
NFWH	NON-FREEZE WALL HYDRANT	
NSD	NOT IN CONTRACT	
OD	OVERFLOW ROOF DRAIN	
OSC	OPEN SITE DRAIN	
PD	PUMPED DISCHARGE	
PDI	PLUMBING & DRAINAGE INSTITUTE	
PRV	PRESSURE REDUCING VALVE	
RD	ROOF DRAIN	
RM	ROOM	
RTU	ROOF TOP UNIT	
SAN	SANITARY	
SFU	SUPPLY FIXTURE UNIT	
ST	STORMWATER	
TP	TRAP PRIMER VALVE	
TPS	TRAP PRIMING STATION	
TW	TEMPERED WATER	
TYP	TYPICAL	
UNO	UNLESS OTHERWISE NOTED	
UR	URINAL	
V	VENT	
VB	VACUUM BREAKER	
VTR	VENT THRU ROOF	
W	WASTE	
W	WITH	
WC	WATER CLOSET	
WCO	WALL CLEANOUT	
WH	WALL HYDRANT	
XR	EXISTING RELOCATED	
LAB/MEDICAL PROJECTS		
O	OXYGEN	
MA	MEDICAL AIR	
LA	LAB AIR	
N	NITROGEN	
LN	LIQUID NITROGEN	
AR	ARGON	
HE	HELIUM	
SG	SPECIAL GAS	
LV	LAB VACUUM	
MV	MEDICAL VACUUM	
CO2	CARBON DIOXIDE	
DI	DEIONIZED WATER	
ROS	REVERSE OSMOSIS SUPPLY	
ROR	REVERSE OSMOSIS RETURN	
LCW	LAB COLD WATER	
LHW	LAB HOT WATER	
LHWR	LAB HOT WATER RECIRCULATING	
VCW	VIVARIUM COLD WATER	
VHW	VIVARIUM HOT WATER	
NP	NON-POTABLE	
SHEET NUMBERING LEGEND		
DISCIPLINE	P111A	SECTOR
"PD" DEMO		
"P" NEW WORK		
PLAN TYPE		FLOOR LEVEL
"1" FLOOR PLAN		"1" LOWEST FLOOR
"2" ENLARGED FLOOR ETC.		"1" NEXT FLOOR LEVEL ETC.
	SYSTEM TYPE	
	"0" FOR UNDERGROUND	
	"1" WATER & GAS	
	"2" DWV & STORM	
	"3" FIRE PROTECTION	

SYMBOL	DESCRIPTION
	DEMOLISH AND REMOVE
	SOIL OR WASTE PIPING
	STORM PIPING
	OVERFLOW STORM PIPING
	NON-POTABLE WATER PIPING
	STORM PIPING BELOW SLAB
	VENT PIPING
	COLD WATER PIPING
	COLD WATER PIPING (SINGLE LINE)
	HOT WATER PIPING
	HOT WATER PIPING (SINGLE LINE)
	HOT WATER RECIRCULATING PIPING
	HOT WATER RECIRCULATING PIPING (SINGLE LINE)
	GAS PIPING
	FIRE PROTECTION PIPING
	PIPE DOWN OR DROP
	PIPE UP OR RISE/FLOOR CLEANOUT
	CAPPED OR PLUGGED PIPE
	UNION
	WALL CLEANOUT
	GATE VALVE
	BALL VALVE
	GLOBE VALVE
	BUTTERFLY VALVE
	GAS COCK
	SHUTOFF VALVE IN RISE OR DROP
	DOUBLE CHECK BACKFLOW PREVENTER
	RPZ BACKFLOW PREVENTER
	CHECK VALVE
	BALANCING VALVE
	OUTSIDE STEM & YOKE VALVE
	PRESSURE GAGE W/ NEEDLE VALVE
	THERMOMETER
	TEMPERATURE & PRESSURE RELIEF VALVE
	VACUUM BREAKER
	WATER HAMMER ARRESTER, LETTER INDICATES PDI
	HOSE BIBB
	WALL HYDRANT
	SOLENOID VALVE
	STRAINER
	PIPE GUIDE/SLEEVE
	PIPE ANCHOR
	RISER/DETAIL DESIGNATION
	FLOW DIRECTION
	CALL OUT DESIGNATION
	FLOOR DRAIN
	METER (GAS/WATER/FIRE)
	PUMP
	PUMP
	KEYNOTE TAG
	LIMIT OF DEMOLITION

DOUBLE SYMBOLS	
	GATE VALVE
	BUTTERFLY VALVE
	3-WAY VALVE
	GAS COCK VALVE
	BALL VALVE
	BALANCE VALVE
	SHUT-OFF VALVE
	BACKFLOW PREVENTER VALVE

										VAY		AA		08/15/2022		Project Number		
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ELECTRIC WATER HEATER SCHEDULE													
ID	LOCATION	STORAGE CAPACITY (GALLONS)	FUEL TYPE	CAPACITY KW	MIN. UNIFORM ENERGY FACTOR	GPH @ 70 DELTA T	INLET TEMP (°F)	OUTLET TEMP (°F)	ELECTRICAL CHARACTERISTICS			BASIS OF DESIGN (OR EQUAL)	REMARKS
									VOLTS	PHASE	HZ		
EW-H-1	MECHANICAL CLOSET	40	ELECTRIC	4.5	0.92	26	70	120	240	1	60	A.O. SMITH ENJ-40	
NOTES:													

SLOPE OF HORIZONTAL DRAINAGE PIPE	
SIZE (INCHES)	MINIMUM SLOPE (INCH PER FOOT)
2-1/2" OR LESS	1/4
3 TO 6	1/8
8 OR LARGER	1/16

TABLE 704.1 OF THE INTERNATIONAL PLUMBING CODE 2018

PIPING MATERIAL SCHEDULE	
PLUMBING SYSTEM	PLUMBING MATERIAL DESCRIPTION
SANITARY DRAIN PIPING (BELOW SLAB)	SOLID CORE SCHEDULE 40 PVC AND FITTINGS
SANITARY DRAIN PIPING (ABOVE SLAB)	SOLID CORE SCHEDULE 40 PVC AND FITTINGS
VENT PIPING	SOLID CORE SCHEDULE 40 PVC AND FITTINGS
DOMESTIC WATER PIPING (ABOVE SLAB)	TYPE "L" COPPER PIPING
DOMESTIC WATER PIPING (BELOW SLAB)	TYPE "K" COPPER PIPING W/ NO JOINTS

[illegible]

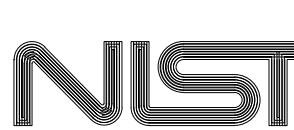
CLEANOUT SCHEDULE		
ID	APPLICATION	BASIC OF DESIGN (OR EQUAL)
CO	FINISHED FLOORS	ZURN Z1400
	CERAMIC & TERRAZO	ZURN Z1400-Z
	UNFINISHED FLOORS	ZURN Z1400-HD
WCO	LINE CLEANOUT	ZURN Z1468
	STACK CLEANOUT	ZURN Z1466
ECO	OUTSIDE AREAS	ZURN Z1400-HD

NOTES

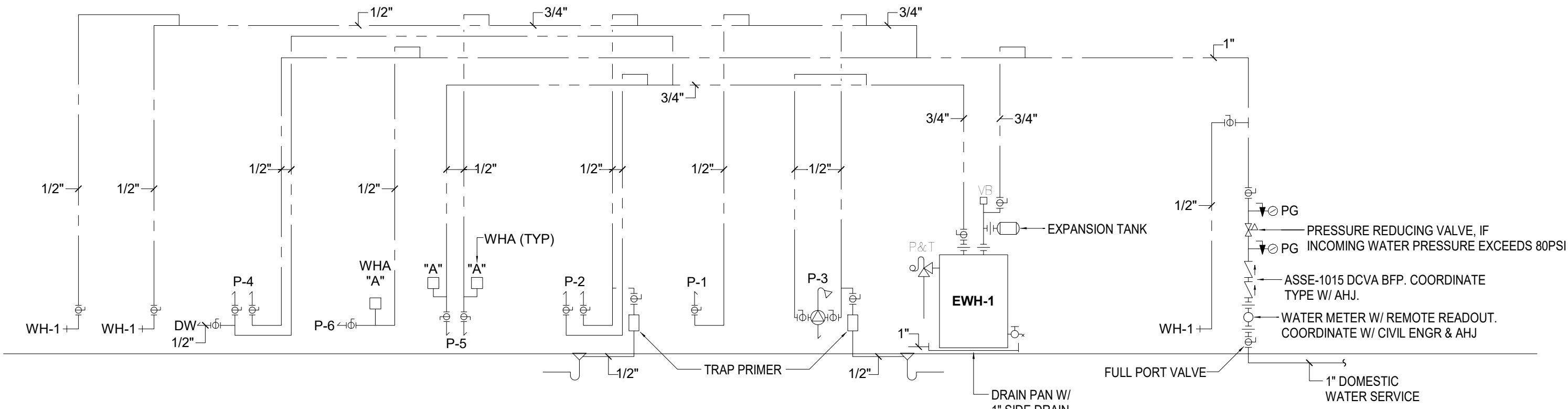
1. INSTALL ECO IN 24x24x4 THICK CONCRETE PAD. TOP OF PAD ELEVATION TO BE 2" ABOVE FINISHED GRADE.
2. PROVIDE ALL CLEANOUTS WITH WIDE ANCHOR FLANGE & CLAMPS DEVICE. ATTACH CLEANOUTS TO MEMBRANE IN FLOOR. WHEN CLEANOUTS ARE LOCATED IN NONMEMBRANE FLOORS, THEY SHALL BE FLASHED IN ACCORDANCE WITH THE SPECIFICATIONS.
3. PROVIDE LINE & STACK TYPE WALL CLEANOUTS WITH ACCESS COVERS.

ID	DESCRIPTION	BASIS OF DESIGN (OR EQUAL)	CONNECTION SIZE				REMARKS
			WASTE	VENT	CW	HW	
P-1	WATER CLOSET	FLOOR MOUNTED, FLUSH TANK, TWO PIECE ELONGATED BOWL, AMERICAN STANDARD 27FA001 020 WITH ELONGATED SOLID PLASTIC SEAT, CLOSED FRONT, 3/8" CHROME PLATED SUPPLY & STOP WAX BOWL RING & WATER CLOSET BOLTS TRIP LEVER TO BE ON THE OPENSIDE, WHITE COLOR	3"	2"	1/2"	---	1.6 GPF MAX OR AS REQUIRED BY THE GOVERNING CODES
P-2	LAVATORY	WHITE VITREOUS CHINA, UNDERMOUNT BOWL 17"X14"X15.5", AMERICAN STANDARD OVAL 0496.221.020 WITH DELTA FAUCET LAHARA COLLECTION, MODEL 526-55MPU-DST DIAMOND COATED CERAMIC CARTRIDGE SINGLE LEVER WITH METALLIC POP-UP DRAIN, 2.2 GPM AT 60PSI, STAINLESS STEEL FINISH, PROVIDE CAST BRASS TRAP & SUPPLIES & STOPS, STAINLESS STEEL FINISH	1-1/2"	1-1/2"	1/2"	1/2"	2.2 GPM MAX OR AS REQUIRED BY THE GOVERNING CODES
P-3	TUB & SHOWER	WHITE CAST IRON, 60"X30"X14", WITH ENAMEL FINISH, KOHLER VILLAGER MODEL K-715 WITH DELTA THERMOSTATICALLY CONTROLLED SHOWER VALVE, STAINLESS STEEL FINISH, SCALD-GUARD, METAL LEVER HANDLE, SHOWER ARM, SHOWER HEAT, DIVERTER VALVE	2"	1-1/2"	1/2"	1/2"	2.5 GPM MAX OR AS REQUIRED BY THE GOVERNING CODES
P-4	SINK	SINGLE BASIN, STAINLESS STEEL SINK, 25"X21"X6" DROP 18 GAUGE WITH SOUND DAMPENING, ELKAY DSE6K12/221 WITH SINGLE HOLE FOR FAUCET & SOAP DISPENSER, KRAUS KPF-16165 FAUCET WITH DUAL FUNCTION PULL-DOWN SPRAYHEAD, BRASS TRAP & SUPPLIES & STOPS WITH STAINLESS STEEL FINISH	2"	1-1/2"	1/2"	1/2"	2.2 GPM MAX OR AS REQUIRED BY THE GOVERNING CODES. PROVIDE A 1/2 HP GARBAGE DISPOSAL
P-5	WASH BOX	WASH BOX OATEY MODEL 38569 OR EQUAL	2"	1-1/2"	1/2"	1/2"	WITH 1/4 TURN VALVES & WATER HAMMER ARRESTOR
P-6	ICE MAKER VALVE BOX	WATER-TITE MODEL #87978 OR EQUAL	-	-	1/2"	-	WITH WATER HAMMER ARRESTOR

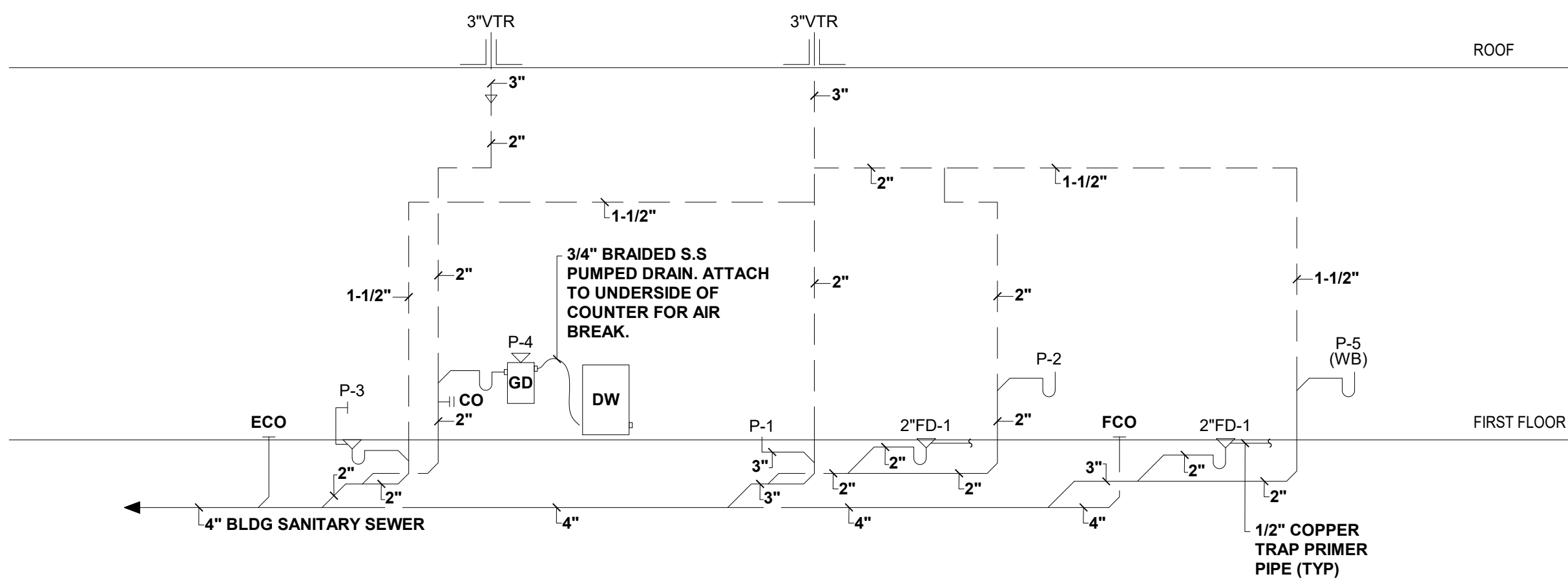
FLOOR DRAIN SCHEDULE					
ID	TYPE	LOCATION/AREA	SIZE	BASIS OF DESIGN (OR EQUAL)	REMARKS
FD-1	FLOOR DRAIN	GENERAL	12X12	ZURN #Z566	W/ 1/2" TRAP PRIMING LINE

REVISION	DESCRIPTION	BY	DATE
			ARCHITECT
			MEP. ENG.
			
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OFFICE OF FACILITIES & PROPERTY MANAGEMENT			
NIST - SINGLE FAMILY DETACHED HOME SINGLE STORY			
PLUMBING SCHEDULES			
P401			
SH. NO. ____ OF ____		DWG. P401	

	
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PLUMBING SCHEDULES	
P401	
SH. NO. ____ OF ____	DWG. P401



1 WATER RISER DIAGRAM
NTS



2 SANITARY RISER DIAGRAM
NTS

NIST

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NIST - SINGLE FAMILY DETACHED HOME
SINGLE STORY

PLUMBING RISER DIAGRAMS

P501

DWG. P501

SH. NO. ____ OF ____

Project Number

08/15/2022

AA

Project Leader

Facilities Engr. Off.

Shed - NIST Fire Marshal/AHJ

Safety

Environmental Management

Maintenance Eng. Off.

Sponsor

Approved By

	PLUMBING GENERAL CONDITIONS
1.	THE INTENT OF THESE DRAWINGS IS TO PROVIDE COMPLETE AND PROPERLY FUNCTIONING PLUMBING SYSTEMS. PROVIDE ALL LABOR AND MATERIAL NECESSARY TO ACHIEVE SUCH ENDS. CONTRACTOR IS OBLIGATED TO EXAMINE PLANS, ANY OBSERVED FAULTS OR AMBIGUITY IN THESE PLANS SHALL BE CALLED TO THE ATTENTION OF THE OWNER'S REPRESENTATIVE IMMEDIATELY, SO THAT THE MATTER MAY BE RESOLVED PRIOR TO SUBMISSION OF BIDS. BY SUBMISSION OF BID, THE CONTRACTOR SHALL ACKNOWLEDGE ACCEPTANCE OF THESE PLANS AS AN ADEQUATE DEFINITION OF THE SCOPE OF WORK AND EXTRA COST CLAIMS BASED ON INADEQUACY OF PLANS WILL NOT BE CONSIDERED.
2.	ALL WORK ON THIS PROJECT SHALL BE INSTALLED IN ACCORDANCE WITH APPLICABLE CODES AND REGULATIONS.
3.	THE PLUMBING CONTRACTOR SHALL COORDINATE THE INSTALLATION OF PLUMBING WORK WITH OTHER TRADES. MINOR DEVIATIONS FROM THE PLANS MAY BE MADE TO AVOID MINOR CONFLICTS. WHEN MAJOR CONFLICTS ARE APPARENT, THE ENGINEER SHALL BE ADVISED IMMEDIATELY, AND AFFECTED WORK SHALL NOT BE INSTALLED UNTIL THE CONFLICT HAS BEEN RESOLVED.
4.	CONTRACTOR SHALL OBTAIN AND PAY FOR PERMITS AND ARRANGE FOR INSPECTIONS BY LOCAL AUTHORITIES HAVING JURISDICTION.
5.	PROVIDE OPENINGS IN BUILDING CONSTRUCTION FOR PASSAGE OF PIPING. DO NOT PENETRATE STRUCTURAL MEMBERS WITHOUT PRIOR APPROVAL OF ARCHITECT AND STRUCTURAL ENGINEER. VERIFY SLAB PENETRATION LOCATIONS PRIOR TO PROCEEDING WITH WORK IN ORDER TO LOCATE OBSTRUCTIONS EMBEDDED IN SLAB.
6.	ALL NECESSARY ALLOWANCES AND PROVISIONS SHALL BE MADE BY THIS CONTRACTOR FOR BEAMS, COLUMNS OR OTHER OBSTRUCTIONS OF THE BUILDING OR THE WORK OF OTHER CONTRACTORS. WHETHER OR NOT SAME IS INDICATED. WHERE NECESSARY TO AVOID OBSTRUCTIONS, ALL DUCTS SHALL BE TRANSFORMED, DIVIDED, OFFSET, RAISED OR LOWERED WITH THE REQUIRED FREE AREA BEING MAINTAINED WHILE MAINTAINING DESIGNING CEILING HEIGHTS.
7.	ALL ROTATING PLUMBING EQUIPMENT SHALL BE MOUNTED ON VIBRATION ISOLATION IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
8.	INSTALL ALL EQUIPMENT IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.
9.	CONTROLS AND ACCESSORIES FOR EQUIPMENT SHALL BE PROVIDED BY EQUIPMENT MANUFACTURER.
10.	PLUMBING CONTRACTOR SHALL THOROUGHLY CLEAN HIS WORK AREA DAILY AND SHALL ALSO REMOVE ALL TRASH AFTER WORK COMPLETION.
11.	THE WORK DETAILED ON THESE PLANS IS BASED ON PREVIOUSLY PREPARED BASE BUILDING DRAWINGS. IF EXISTING FIELD CONDITIONS ARE DIFFERENT THAN SHOWN, THE CONTRACTOR SHALL ADVISE THE ENGINEER OF DISCREPANCIES WHICH WILL AFFECT THE PROPOSED WORK PRIOR TO BEGINNING WORK.
12.	SYMBOLS SHOWN ON SCHEDULES DEFINE TYPE OF EQUIPMENT ONLY. CONTRACTOR IS RESPONSIBLE FOR RESEARCHING DRAWINGS FOR EXACT QUANTITIES REQUIRED OF EACH TYPE.
13.	CONTRACTOR SHALL VERIFY THAT ALL EQUIPMENT, PIPING, ETC., SHALL BE FREE FROM INTERFERENCE WITH OTHER DISCIPLINES. IF CONFLICTS EXIST, THE CONTRACTOR SHALL IMMEDIATELY CONTACT THE CONSTRUCTION MANAGER. WHEN THE WORK OF VARIOUS TRADES WILL BE INSTALLED IN CLOSE PROXIMITY TO ONE ANOTHER, OR WHERE THERE IS EVIDENCE THAT THE WORK OF ONE TRADE WILL INTERFERE WITH WORK OF OTHER TRADES, THE CONTRACTOR SHALL ASSIST IN WORKING OUT SPACE CONDITIONS TO MAKE A SATISFACTORY ADJUSTMENT. IF THE CONTRACTOR ALLOWS ONE TRADE TO INSTALL HIS WORK BEFORE COORDINATING WITH WORK OF OTHER TRADES, THE CONTRACTOR SHALL MAKE NECESSARY CHANGES TO CORRECT THE CONDITION WITHOUT EXTRA CHARGE.
14.	THE CONTRACTOR SHALL LOCATE ALL EQUIPMENT WHICH MUST BE SERVICED, OPERATED, MAINTAINED IN FULLY ACCESSIBLE POSITION. EQUIPMENT SHALL INCLUDE, BUT NOT BE LIMITED TO, VALVES, TRAPS, CLEANOUTS, MOTORS, CONTROLLERS, DRAIN POINTS, ETC. IF REQUIRED FOR ACCESSIBILITY, FURNISH ACCESS DOORS FOR THIS PURPOSE. MINOR DEVIATIONS FROM DRAWINGS MAY BE MADE, ALLOW FOR BETTER ACCESSIBILITY, AND ANY CHANGE FOR THAT PURPOSE SHALL BE APPROVED.
15.	ALL EQUIPMENT WHICH REQUIRES INTERFACE WITH THE ELECTRICAL TRADE, SUCH AS CONTROLLING DEVICES, DISCONNECT SWITCHES, MOTORS, ETC., SHALL BE COORDINATED WITH THE ELECTRICAL TRADE.
16.	ALL WORK TO BE GUARANTEED FOR ONE YEAR AGAINST LABOR AND MATERIALS. ANY DEFECTIVE MATERIALS OR WORKMANSHIP, AS WELL AS DAMAGE TO THE YIELD OF ALL TRADES RESULTING FROM SUCH WORK, SHALL BE REPLACED AT THE CONTRACTOR'S EXPENSE DURING THE DURATION OF THE GUARANTEE PERIOD. TIME FOR THIS GUARANTEE SHALL BEGIN FROM THE DATE OF ACCEPTANCE OF THE COMPLETE WORK BY THE OWNER OR HIS APPOINTED REPRESENTATIVE. NOTE: THESE GUARANTEES SHALL BE SUBMITTED TO THE CO FOR RECORD PURPOSES.
17.	AS USED IN THE DRAWINGS FOR PLUMBING WORK, CERTAIN NON-TECHNICAL WORDS SHALL BE UNDERSTOOD TO HAVE SPECIFIC MEANINGS AS FOLLOWS REGARDLESS OF INDICATIONS TO THE CONTRARY IN THE GENERAL CONDITION OR OTHER DOCUMENTS GOVERNING THE MECHANICAL WORK. "FURNISH" - PURCHASE AND DELIVER TO THE PROJECT SITE COMPLETE WITH EVERY NECESSARY APPURTENANCE AND ALL SUPPORT. ALL PART OF THE PURCHASED WORK, PURCHASING SHALL INCLUDE PAYMENT OF ALL SALES TAXES AND OTHER SURCHARGES AS WELL BE REQUIRED TO ASSURE THAT PURCHASED ITEMS ARE FREE OF ALL LIENS, CLAIMS OR ENCUMBRANCES. "INSTALL" - LOAD AT THE DELIVERY POINT AT THE SITE AND PERFORM EVERY OPERATION NECESSARY TO ESTABLISH SECURE MOUNTING AND CORRECT OPERATION AT THE PROPER LOCATION IN THE PROJECT, ALL AS PART OF THE MECHANICAL WORK. "PROVIDE" - "FURNISH" AND "INSTALL". "NEW" - MANUFACTURED WITHIN THE PAST TWO YEARS AND NEVER BEFORE USED.

PLUMBING GENERAL NOTES

- BEFORE COMMENCEMENT OF WORK, CONTRACTOR SHALL VERIFY EXACT LOCATIONS, ELEVATIONS AND CHARACTERISTICS OF UTILITIES AND PIPING AND SHALL IMMEDIATELY NOTIFY THE ARCHITECT/ENGINEER OF ANY DISCREPANCIES.
- THE ELECTRICAL CONTRACTOR IS REQUIRED TO PROVIDE POWER FOR ALL PLUMBING EQUIPMENT. DURING BID PERIOD AND AGAIN PRIOR TO EQUIPMENT ORDERS, THE GENERAL CONTRACTOR AND THE PLUMBING CONTRACTOR SHALL COORDINATE WITH THE ELECTRICAL CONTRACTOR TO VERIFY THAT ALL EQUIPMENT ARE POWERED, AND PROPER VOLTAGE/WIRING/CAPACITY IS PROVIDED. IN THE EVENT OF REQUIREMENTS AMBIGUITY OR CONFLICT BETWEEN TRADES, THE CONTRACTOR IS REQUIRED TO GENERATE AN RFPI DETERMINING THE ISSUE AND REQUESTING AMBIFICATION. FAILURE TO DO SO WILL NEGATE HIS RIGHT FOR A CHANGE ORDER OR REQUEST FOR ADDITIONAL FUNDS DURING CONSTRUCTION RELATING TO THIS ISSUE.
- DRAWINGS ARE IN PART DIAGRAMMATIC, INTENDED TO CONVEY SCOPE OF WORK, AND TO INDICATE THE GENERAL LOCATIONS OF EQUIPMENT, PIPING, ETC. CONTRACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND LAYOUT HIS WORK ACCORDING TO THE FOLLOWING GUIDELINES:
 - CONTRACTOR SHALL BE RESPONSIBLE FOR DETERMINING THE EXACT LOCATIONS FOR EQUIPMENT AND ROUGH INS AND THE EXACT ROUTING OF PIPING PRIOR TO CONSTRUCTION AS TO BEST FIT THE LAYOUT OF THE WORK. COORDINATE FINAL LAYOUT WITH ALL TRADES.
 - WHERE OFFSETS IN PIPING ARE REQUIRED TO COORDINATE WORK OF OTHER TRADES, STRUCTURE, PIPING, CONDUITS, DUCTWORK, ETC. OR TO MAINTAIN CEILING HEIGHTS, THEY SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
 - UNLESS OTHERWISE NOTED, ALL PIPING TO BE ROUTED CONCEALED IN WALLS, CHASES OR ABOVE CEILING. WATER PIPING SHALL NOT BE ROUTED IN EXTERIOR WALLS. ROUTE ALL PIPING AS HIGH AS POSSIBLE AND ALONG WALLS TO MAXIMIZE SPACE AVAILABLE FOR OTHER TRADES.
 - COORDINATE ROUTING OF PIPING TO MAINTAIN ACCESS TO FILTERS, MOTORS, ELECTRICAL EQUIPMENT, AND CONTROLS. IN NO CASE SHALL PIPING PASS DIRECTLY OVER ELECTRICAL PANELS OR DISCONNECTS OR RESTRICT ACCESS TO ANY ELECTRICAL EQUIPMENT INCLUDING JUNCTION BOXES.
 - ALL EXISTING PIPING SHOWN IS INTENDED TO INDICATE APPROXIMATE SIZE, NUMBER AND LOCATION OF PIPING BRANCHES FOR BIDDING PURPOSES ONLY. CONTRACTOR TO VERIFY EXACT SIZE AND CONFIGURATION PRIOR TO CONSTRUCTION.
 - VERIFY EXACT SIZE AND LOCATION OF ALL EXISTING SYSTEMS REQUIRING CONNECTION TO NEW PIPING PRIOR TO COMMENCING WORK.
- UNLESS OTHERWISE NOTED, SANITARY WASTE PIPING SHOWN IS BELOW FLOOR AND ALL OTHER PIPING IS OVERHEAD, ABOVE CEILING, DOMESTIC HOT, COLD AND RE-CIRC. WATER PIPING SHALL BE INSTALLED ON WARM SIDE OF INSULATION.
- ALL PLUMBING WORK SHALL BE DONE IN ACCORDANCE WITH 2018 INTERNATIONAL PLUMBING CODE (IPC) & ALL LOCAL AMENDMENTS, CODES AND REGULATIONS.
- DURING THE PROGRESS OF THE WORK, MAINTAIN AN ACCURATE RECORD OF ALL CHANGES MADE IN THE PLUMBING SYSTEMS. THE RECORD DRAWING SHALL SHOW CHANGES IN MANUFACTURER (WITH NUMBERS AND NAMES), MATERIALS, SIZES, LOCATING AND HOOK-UP POINTS. AS-BUILTS SHALL BE GIVEN TO [] OWNER [] AT COMPLETION OF JOB.
- UPON COMPLETION OF JOB, THIS CONTRACTOR SHALL INSPECT ALL EXPOSED PORTIONS OF THE PLUMBING INSTALLATION AND COMPLETELY REMOVE ALL EXPOSED LABELS, SOIL, MARKING AND FOREIGN MATERIAL EXCEPT PRODUCT LABELS AND THOSE REQUIRED BY LAW TO REMAIN.
- ALL INSTALLATION OF ACCESSIBLE PLUMBING FIXTURES SHALL BE PERFORMED IN ACCORDANCE WITH THE REQUIREMENTS OF THE STATE ACCESSIBILITY STANDARDS. REFER TO ARCHITECTURAL PLANS FOR SPECIFIC INSTALLATION REQUIREMENTS. LOCATE WATER CLOSET FLUSH VALVE LEVER BETWEEN TOILET AND LAVATORY. COORDINATE LOCATION OF FLUSH VALVE LEVER HANDLE WITH GRAB BAR. COORDINATE WITH ARCHITECTURAL DRAWINGS.
- ALL VENT AND FLUE OUTLETS SHALL BE 10'-0" MINIMUM HORIZONTALLY AND 3'-0" MINIMUM VERTICALLY FROM ANY FRESH AIR INTAKES.
- PROVIDE WATER HAMMER ARRESTORS AT ALL PLUMBING FIXTURES OR BATTERY OF FIXTURES WITH QUICK-CLOSING VALVES. WATER HAMMER ARRESTORS SHALL BE SIZED & INSTALLED PER PDI STANDARDS. AIR CHAMBERS SHALL NOT BE CONSIDERED AN EQUAL TO WATER HAMMER ARRESTORS AND SHALL NOT BE INSTALLED.
- PROVIDE BALANCING VALVES WITH THREADED CONNECTIONS IN HOT WATER RETURN PIPING, AND BALANCE SYSTEM FOR PROPER OPERATION.
- COORDINATE WITH ARCHITECTURAL DRAWINGS BEFORE ROUGHING IN PLUMBING.
- ALL OPENINGS IN CEILINGS AND PLENUM WALLS FOR PLUMBING SHALL BE SEALED AIR TIGHT AND PROTECTED WITH FIRESTOP.
- SEE SITE PLAN FOR EXTENT OF ALL PIPING LEAVING AND ENTERING BUILDING.
- ALL HOSE BIBS SHALL BE MOUNTED 18" ABOVE FINISHED FLOOR, UNLESS OTHERWISE SPECIFIED.
- ALL WALL HYDRANTS SHALL BE MOUNTED 24" ABOVE FINISHED GRADE UNLESS OTHERWISE SPECIFIED.
- MAKE PROPER HW, CW, RE-CIRC., WASTE, AND VENT CONNECTIONS TO ALL FIXTURES AND EQUIPMENT EVEN THOUGH ALL BRANCH MAIN, ELBOWS AND CONNECTIONS ARE NOT SHOWN.
- CLEANOUTS SHALL BE PROVIDED NEAR BASE OF EACH VERTICAL WASTE OR SOLID STACK. PROVIDE 18" MINIMUM CLEARANCE FOR ACCESS.
- PROVIDE VALVES AND UNIONS WHEN CONNECTING PIPING TO EQUIPMENT. LOCATE VALVES FOR EASY ACCESS AND SERVICING OF EQUIPMENT. PROVIDE ACCESS PANELS TO ALL SHUT-OFF VALVES.
- PRESSURE TEST ALL NEW WATER PIPES, MINIMUM OF 4 HOURS BEFORE APPLYING PIPE INSULATION. ALL TESTS SHALL BE IN COMPLIANCE WITH REQUIREMENTS OF THE LOCAL WATER AUTHORITY.
- IN THE EVENT OF CONFLICT BETWEEN VARIOUS SECTIONS OF CONSTRUCTION DOCUMENTS, THE MOST STRINGENT OF THE OPTIONS SHALL APPLY FOR BIDDING PURPOSE. AFTER AWARD OF CONTRACT BUT BEFORE STARTING DESIGN/CONSTRUCTION, CONTRACTOR SHALL REQUEST CLARIFICATIONS WITH BID ON ANY CONFLICTING INFORMATION FROM THE OWNER.
- PROVIDE DIELECTRIC FITTINGS OTHER THAN DIELECTRIC UNIONS WHEN CONNECTING DISSIMILAR MATERIAL PIPING. DIELECTRIC UNIONS NOT PERMITTED.
- ALL CONCRETE FLOORS, SLABS, BRICK AND BLOCK WALL PENETRATIONS SHALL BE SLEEVED AND FIRE STOPPED.
- PROVIDE BAL VALVE FOR ALL CW AND HW BRANCH TAKE-OFFS.
- ALL EXTERIOR ROOF PENETRATIONS SHALL BE SEALED WEATHERTIGHT.
- SLOPES AND INVERT ELEVATIONS OF ALL INTERIOR PIPING SHALL BE ESTABLISHED BEFORE ANY PIPING IS INSTALLED IN ORDER THAT PROPER SLOPES WILL BE MAINTAINED. ALL PIPING SHALL BE LOCATED AND DETERMINED WHERE TO BE RUN TO AVOID CONFLICT WITH OTHER TRADES.
- ALL PIPING SHALL BE PROVIDED WITH SECTIONAL SHUT-OFF VALVES AND ACCESS PANELS FOR ALL HIDDEN VALVES. PROVIDE ISOLATION VALVES FOR ALL PLUMBING FIXTURES.
- ALL WORK SHALL BE GUARANTEED AGAINST DEFECTS, LEAKS, LACK OF PROPER SYSTEM PERFORMANCE OR NON-OPERATION FOR A PERIOD OF ONE YEAR AFTER DATED OF FINAL ACCEPTANCE OF WORK BY OWNER.
- ALL WORK SHALL BE COORDINATED WITH ALL TRADES PRIOR TO INSTALLATION.
- ALL EXPOSED PIPING TO FIXTURES SHALL BE HEAVILY CHROME-PLATED.
- MAKE FINAL CONNECTIONS AFTER FIXTURES AND EQUIPMENT ARE SET IN PLACE. ALL REQUIRED TRIM, FAUCETS AND MINOR ITEMS NECESSARY FOR A COMPLETE FUNCTIONING INSTALLATION, SUCH AS STRAINERS, TRAPS, TAILPIECES, ESCUTCHEONS, ETC., SHALL BE FURNISHED AND INSTALLED BY THE CONTRACTOR IN ACCORDANCE WITH EQUIPMENT REQUIREMENTS. ALL FAUCETS TO BE HEAVILY CHROME-PLATED.
- SLEEVES THROUGH CMU BLOCKS WALLS AND/OR MASONRY FLOORS SHALL BE SCHEDULE 40 BLACK STEEL PIPE. SLEEVES THRU FLOORS SHALL STAND PROUD OF FLOOR SLAB BY 1/2". SLEEVES THROUGH SHEET ROCK WALL SHALL BE 18" GAUGE GALVANIZED SHEET METAL. THE SPACE BETWEEN THE PIPE AND SLEEVE SHALL BE PACKED WITH "3M" EXPANDABLE FOAM FIRE-STOP MATERIAL. TO ASSURE A RATING EQUAL TO THE PENETRATED WALL OR FLOOR.
- HANGERS SHALL BE OF SIMILAR MATERIAL AS PIPING TO AVOID GALVANIC ACTION IN ACCORDANCE WITH MSS-SP-69, AND LOCATED AT INTERVALS NOT TO EXCEED 6'-0" C. WHERE "HUBLESS" PIPE IS BEING USED, PROVIDE HANGERS ON EITHER SIDE OF THE JOINTS AS RECOMMENDED BY EXCIP BULLETIN 310. FOR INSULATED PIPING, INSULATION SHALL EXTEND CONTINUOUSLY THRU HANGER WITH OUT INTERRUPTION. PROVIDE SADDLE TO PROTECT INSULATION FROM COMPRESSION.
- SOIL, WASTE AND VENT PIPING SHALL BE INSTALLED IN ACCORDANCE WITH CODE. HORIZONTAL DRAINAGE PIPING SHALL SLOPE PER 2018 IPC CODE, UNLESS OTHERWISE INDICATED. DOMESTIC WASTE SHALL BE SLOPED TO DRAIN POINTS.
- SUBMIT OPERATING AND MAINTENANCE MANUALS AND WARRANTY CERTIFICATES TO OWNER.
- THE CONTRACTOR SHALL NOT CORE DRILL CONCRETE SLABS OR PRE-CAST CONCRETE PANELS OR DECK WITHOUT THE KNOWLEDGE AND WRITTEN CONSENT OF THE STRUCTURAL ENGINEER AND THE BUILDING OWNER AND SHALL X-RAY PRIOR TO ANY CORE DRILLING, X-RAYING AND CORE DRILLING SLABS SHALL BE SCHEDULED WITH THE BUILDING OWNER. INCLUDE COSTS OF X-RAYING IN BID.
- FINAL ACCEPTANCE OF PLUMBING SYSTEMS BY THE USER/OWNER, SHALL BE CONTINGENT TO COMPLIANCE WITH ALL COMMISSIONING REQUIREMENTS AND TESTS PRESCRIBED IN THE PROJECT DESIGN DOCUMENTS/SPECIFICATIONS AND APPROVAL BY THE CxA AGENT OF THE PROJECT.
- UPON COMPLETION OF THE PROJECT, ALL DOMESTIC WATER PIPING SHALL BE SANITIZED IN ACCORDANCE WITH THE REQUIREMENTS OF THE AUTHORITY HAVING JURISDICTION.
- WHEN NATURAL GAS IS AVAILABLE, USE NATURAL GAS RANGES/OVENS & FURNACES IN SINGLE FAMILY & MULTI-FAMILY BUILDINGS AS AN OPTION, IN-LIEU OF ELECTRIC.

ABBREVIATIONS		
ABV	ABOVE	
AD	AREA DRAIN	
AF	ABOVE FINISH FLOOR	
AFG	ABOVE FINISH GRADE	
AP	ACCESS PANEL	
APPROX	APPROXIMATE	
AV	ACID VENT	
AW	ACID WASTE	
BAS	BUILDING AUTOMATION SYSTEM	
BFF	BELOW FINISH FLOOR	
BFP	BACKFLOW PREVENTER	
BLDG	BUILDING	
BT	BATHTUB	
CA	COMPRESSED AIR	
CD	CONDENSATE	
CLG	CEILING	
CO	COMPANY, CLEANOUT	
CW	COLD WATER	
DFU	DRAINAGE FIXTURE UNITS	
DIA	DIAMETER	
DN	DOWN	
DS	DOWNSPOUT	
DWG	DRAWING	
EW	COLD WATER	
DISP	DISPOSER	
DWV	DRAINAGE WASTE & VENT	
EACH	EACH	
ECO	EXTERIOR CLEANOUT	
ETR	EXISTING TO REMAIN	
EW	EMERGENCY EYEWASH	
ES	EMERGENCY SHOWER	
EWC	ELECTRIC WATER COOLER	
EW	ELECTRIC WATER HEATER	
EX	EXISTING	
FE	FLOOR DRAIN	
FPE	FINISHED FLOOR ELEVATION	
FIXT	FIXTURE	
FLR	FLOOR	
FM	FORCED MAIN	
FS	FLOOR SINK	
G	NATURAL GAS	
GCO	GRADE CLEANOUT	
GD	GROUNDWATER DRAIN TILE	
GPF	GALLONS PER FLUSH	
GPM	GALLONS PER MINUTE	
GW	GAS WATER HEATER	
HB	HOSE BIB	
HW	HOT WATER	
HWR	HOT WATER RECIRCULATING	
IE	INVERT ELEVATION	
IND	INDIRECT WASTE	
IPC	INTERNATIONAL PLUMBING CODE	
LAV	LAVATORY	
MAU	MAKE-UP AIR UNIT	
MAX	MAXIMUM	
MB	MOP BASIN	
MBH	BRITISH THERMAL UNITS (x1000)	
MIN	MINIMUM	
MISC	MISCELLANEOUS	
NFVH	NON-FREEZE WALL HYDRANT	
NC	NOT IN CONTRACT	
OD	OVERFLOW ROOF DRAIN	
OSD	OPEN SITE DRAIN	
PD	PUMPED DISCHARGE	
PDI	PLUMBING & DRAINAGE INSTITUTE	
PRV	PRESSURE REDUCING VALVE	
RD	ROOF DRAIN	
RM	ROOM	
RTU	ROOF TOP UNIT	
SAN	SANITARY	
SFU	SUPPLY FIXTURE UNIT	
ST	STORMWATER	
TP	TRAP PRIMER VALVE	
TPS	TRAP PRIMING STATION	
TW	TEMPERED WATER	
TV	TYPICAL	
UNO	UNLESS OTHERWISE NOTED	
UR	URINAL	
V	VENT	
VB	VACUUM BREAKER	
VTR	VENT THRU ROOF	
W	WASTE	
WI	WITH	
WC	WATER CLOSET	
WCO	WALL CLEANOUT	
WH	WALL HYDRANT	
XR	EXISTING RELOCATED	
LAB/MEDICAL PROJECTS		
O	OXYGEN	
MA	MEDICAL AIR	
LA	LAB AIR	
N	NITROGEN	
LN	LIQUID NITROGEN	
AR	ARGON	
HE	HELIUM	
SG	SPECIAL GAS	
MV	MEDICAL VACUUM	
CO2	CARBON DIOXIDE	
OJ	DEIONIZED WATER	
ROS	REVERSE OSMOSIS SUPPLY	
ROR	REVERSE OSMOSIS RETURN	
LCW	LAB COLD WATER	
LHW	LAB HOT WATER	
LHWR	LAB HOT WATER RECIRCULATING	
VCW	VIVARIUM COLD WATER	
VHW	VIVARIUM HOT WATER	
NP	NON-POTABLE	
SHEET NUMBERING LEGEND		
DISCIPLINE	P111A	SECTOR
"D" DEMO		
"P" NEW WORK		
PLAN TYPE		FLOOR LEVEL
"1" FLOOR PLAN		"0" LOWEST FLOOR
"2" ENLARGED		"1" NEXT FLOOR
FLOOR ETC.		LEVEL ETC.
	SYSTEM TYPE	
	"0" FOR UNDERGROUND	
	"1" WATER & GAS	
	"2" DWV & STORM	
	"3" FIRE PROTECTION	

PLUMBING SYMBOLS	
SYMBOL	DESCRIPTION
	DEMOLISH AND REMOVE
	SOIL OR WASTE PIPING
	STORM PIPING
	OVERFLOW STORM PIPING
	NON-POTABLE WATER PIPING
	STORM PIPING BELOW SLAB
	VENT PIPING
	COLD WATER PIPING
	COLD WATER PIPING (SINGLE LINE)
	HOT WATER PIPING
	HOT WATER PIPING (SINGLE LINE)
	HOT WATER RECRULATING PIPING
	HOT WATER RECRULATING PIPING (SINGLE LINE)
	GAS PIPING
	FIRE PROTECTION PIPING
	PIPE DOWN OR DROP
	PIPE UP OR RISE/FLOOR CLEANOUT
	CAPPED OR PLUGGED PIPE
	UNION
	WALL CLEANOUT
	GATE VALVE
	BALL VALVE
	GLOBE VALVE
	BUTTERFLY VALVE
	GAS COCK
	SHUTOFF VALVE IN RISE OR DROP
	DOUBLE CHECK BACKFLOW PREVENTER
	RPZ BACKFLOW PREVENTER
	CHECK VALVE
	BALANCING VALVE
	OUTSIDE STEM & YOKE VALVE
	PRESSURE GAGE W/ NEEDLE VALVE
	THERMOMETER
	TEMPERATURE & PRESSURE RELIEF VALVE
	VACUUM BREAKER
	WATER HAMMER ARRESTER, LETTER INDICATES PDI
	HOSE BIBB
	WALL HYDRANT
	SOLENOID VALVE
	STRAINER
	PIPE GUIDE/SLEEVE
	PIPE ANCHOR
	FLEXIBLE CONNECTION
	FLOW DIRECTION
	RISER/DRAIN DESIGNATION
	FLOOR DRAIN
	CALL OUT DESIGNATION
	METER (GAS/WATER/FIRE)
	PUMP
	PUMP
	KEYNOTE TAG
	LIMIT OF DEMOLITION

DOUBLE SYMBOLS

	GATE VALVE
	BUTTERFLY VALVE
	3-WAY VALVE
	GAS COCK VALVE
	BALL VALVE
	BALANCE VALVE
	SHUT-OFF VALVE
	BACKFLOW PREVENNTER VALVE

REVISION	DESCRIPTION	BY	DATE
		ARCHITECT	
		MEP. ENG.	
			
FOR OFFICIAL USE ONLY			
U.S. DEPARTMENT OF COMMERCE			
OFFICE OF FACILITIES & PROPERTY MANAGEMENT			
NIST – SINGLE FAMILY DETACHED HOME MULTI STORY			
PLUMBING COVERSHEET			
P002			
SH. NO. ____ OF ____		DWG. P002	